UNIVERSITY OF UYO

DEPARTMENT OF MATHEMATICS

MTH 111 TEST FIRST SEMESTER 2020/2021 SESSION

Name of Student:_____

Reg. No._____ **GROUP**_____ **SERIAL No.**_____

Department:_____ **Signature**_____

PHYSICAL SCIENCES

Fill the spaces above first. Then answer ALL the Questions by writing the correct answer in the space provided. No phones, no calculators are allowed. Time allowed: 40 minutes

1. Solve the equation $2^{2x}.9^{x} = 36$

2. Let α, β be the roots of the equation $3x^2 - 5x - 1 = 0$, find the value of $\alpha^2 + \beta^2$.

3. Solve $|3x + 1| < ¼$

4. Find the fifth term in the expansion of $(a - b)^{10}$.

5. Find, without tables, the square root of $91 + 40\sqrt{3}$

6. For what values of λ does the equation $(\lambda - 1)x^2 + 4x + \lambda + 2 = 0$ have real roots?

7. Solve the equation for x, $\log_6(2x+1) = \log_{\sqrt{6}}(2x-1)$

8. Find the values of p and q given that $x^4 + x^3 - 7x^2 + px + q$ is divisible by $(x-1)$ and $(x-2)$,

9. Let $A = \{x : 2x^2 - 3x + 1 = 0\}$ *and* $B = \{x : x^2 - 5x + 4 = 0\}$ find $(A \cup B) \times (A \cap B)$

10. Express as a sum of partial fractions $\dfrac{3x^2 - x}{(x-2)(x^2+1)}$